Hospital Emergency Room Dashboard

Analyst Summary Report | Period: 2023-2024 (patient records from April 2023 to October 2024)

1. Executive Summary

Across 2023-2024 the emergency room handled **9,216 patients** (about 485 per month) with an average wait time of **35.26 minutes** and a patient satisfaction score of **4.99 out of 10**. Half of all patients (**50.0%**) were admitted. The key issue is timeliness: **59.3% of patients were attended with a delay**, and the delay rate rose from 58.0% in 2023 to 60.5% in 2024. Demand is steady and evenly spread across age groups, genders, weekdays and hours, and 59% of patients needed no onward referral. The dashboard's monthly view (2024 / January) shows 513 patients, a 36.32-minute wait and 4.96 satisfaction, which is slightly busier and slower than the 19-month average.

2. Key Performance Indicators (2023-2024)

9,216	35.26	4.99	50.0%	59.3%
Total Patients	Avg Wait Time (min)	Satisfaction (0-10)	Admission Rate	Delay Rate

Satisfaction is based on 2,517 survey responses (27% of patients). Wait time is in minutes.

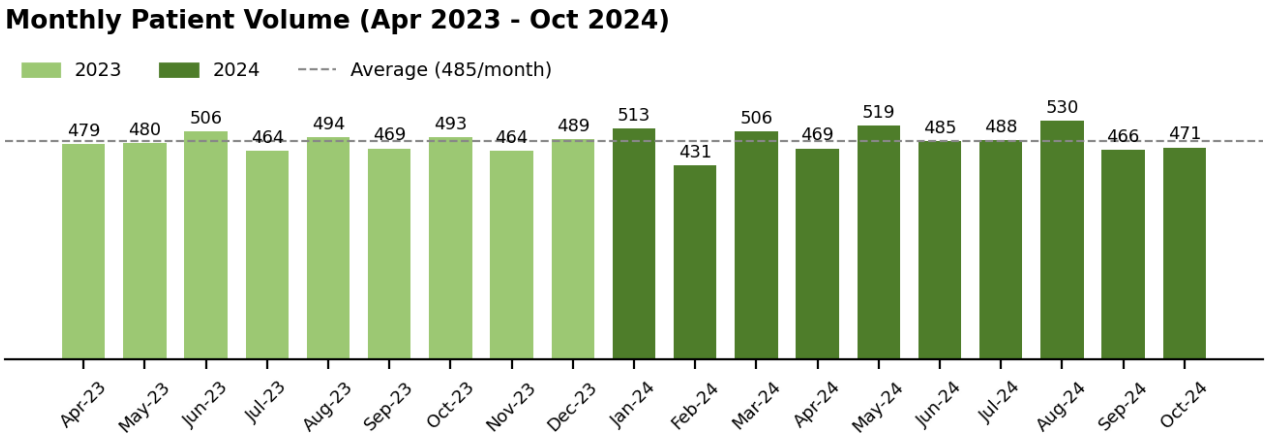
3. Year-over-Year Comparison

Metric	2023 (Apr-Dec)	2024 (Jan-Oct)	Change
Patients (total)	4,338	4,878	-
Avg patients per month	482.0	487.8	+1.2%
Avg wait time (min)	35.03	35.47	+0.44
Satisfaction (0-10)	5.03	4.96	-0.06
Admission rate	49.7%	50.3%	+0.6 pts
Delay rate	58.0%	60.5%	+2.5 pts

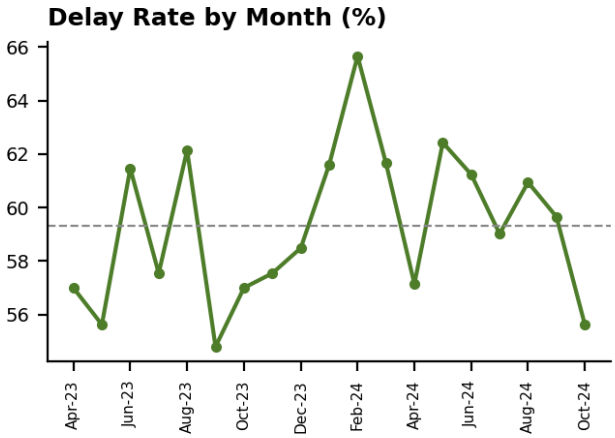
2023 covers nine months and 2024 covers ten, so monthly averages are used for a like-for-like volume comparison.

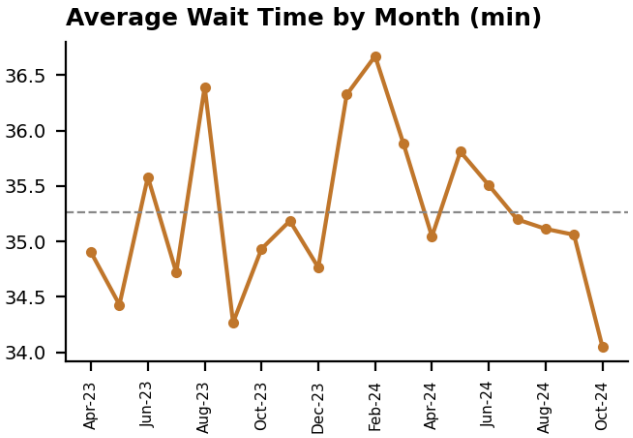
Volume grew modestly (+1.2% per month), while performance moved the wrong way: the delay rate is 2.5 points higher, average wait is 0.44 minutes longer and satisfaction slipped by 0.06 points. Changes are small but consistent in direction.

4. Monthly Trends



- Busiest month: **Aug-24 (530 patients)**. Quietest: **Feb-24 (431)**, which is also the shortest month.





- The delay rate peaked at **65.7% in Feb-24**, the month with the lowest patient volume. Monthly volume and delay rate are only weakly related (correlation 0.21), so delays look driven by process and staffing rather than volume.
- Average wait time stays in a narrow band (34.1 to 36.7 minutes) with no sharp spikes.

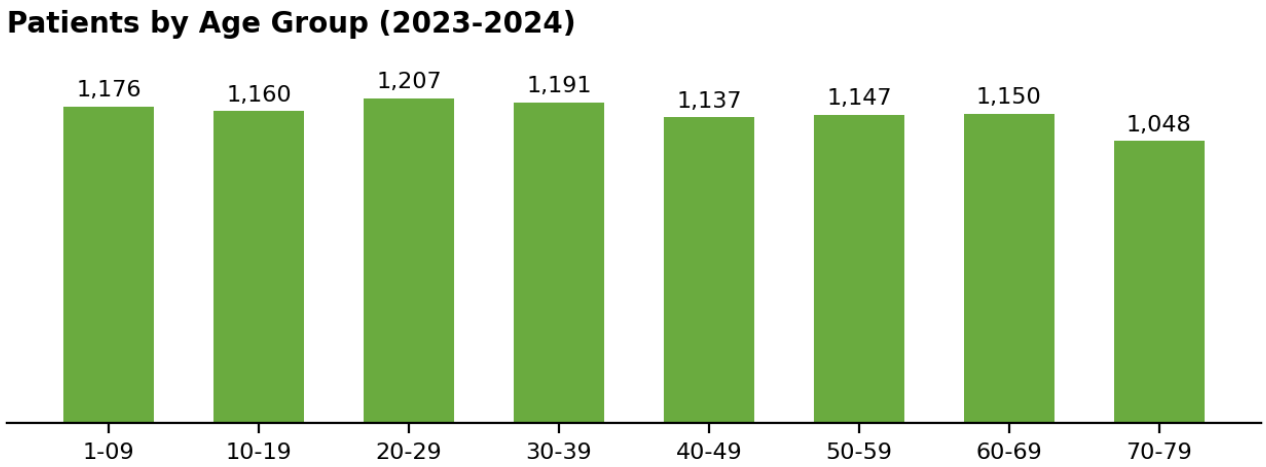
5. Detailed Findings

5.1 Admission, Attendance and Gender

Admission	Patients	Attend Status	Patients	Gender	Patients	Share
Admitted	4,612	Delay	5,467	Male	4,705	51.1%
Not Admitted	4,604	On Time	3,749	Female	4,487	48.7%
				Not classified	24	0.3%

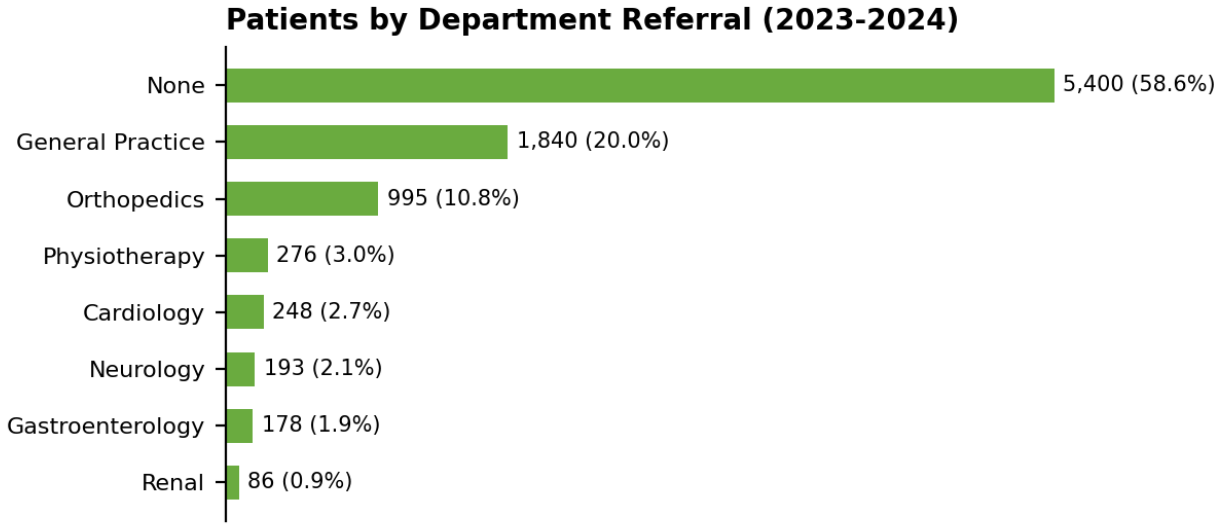
- Admissions and discharges are split almost evenly (50.0% admitted) and stay within about 49-52% across every age group and referral type.
- In the data, a patient is classed as delayed when the wait exceeds 30 minutes. Delayed patients waited **45.7 minutes on average versus 20.1 minutes** for on-time patients.
- Satisfaction barely differs between delayed (4.96) and on-time (5.04) patients, so waiting time alone does not explain the scores.
- Gender is balanced: male 51.1% and female 48.7%, with a very small unclassified group.

5.2 Age Group Distribution



- Patients are spread evenly: every age group holds between 1,048 and 1,207 patients (11.4% to 13.1%).
- The **20-29 group is the largest (1,207)** and **70-79 is the smallest (1,048)**. Children and teens (1-19) account for 25.3% and adults aged 60+ for 23.8%.
- Because no single age band dominates, ER capacity and equipment need to cover all ages rather than one core group.

5.3 Department Referrals



- **5,400 patients (58.6%) had no referral**; the remaining 3,816 (41.4%) were referred onward.
- **General Practice (1,840) and Orthopedics (995)** make up 74% of all referrals.
- Physiotherapy (276), Cardiology (248), Neurology (193), Gastroenterology (178) and Renal (86) are low-volume pathways.
- Demand is steady through the week (1,260 to 1,377 patients per weekday) and across the day (353 to 434 per hour), with no pronounced peak.

6. Insights and Recommendations

- **Reduce delays (priority 1).** Nearly 6 in 10 patients are delayed, and the rate is rising. Set a delay-rate target (for example below 55%) and review triage, staffing and bed turnover.
- **Investigate high-delay months.** Feb-24 had the highest delay rate (65.7%) despite the lowest volume, pointing to process or staffing causes that should be reviewed month by month.
- **Create a primary-care fast track.** General Practice receives 48% of referrals and 20% of all patients; an urgent-care stream could free ER capacity.
- **Staff evenly across the week and day.** With no clear peak by weekday or hour, rosters should provide consistent cover rather than concentrating resources in one window.
- **Strengthen satisfaction tracking.** Only 27% of patients give a score, and scores do not change with delay. Wider survey coverage would give a more reliable picture.
- **Extend the dashboard.** Add year-over-year views, delay-rate trend lines and targets next to each KPI so changes are visible at a glance.

7. Conclusion

Over 2023-2024 the emergency room operated at a stable, high volume of about 485 patients a month, with a balanced patient mix, an even admission split and an average wait of 35.3 minutes. Satisfaction is moderate at 4.99 out of 10 and has not improved. The main concern is timeliness: **59.3% of patients were attended late**, the share is trending up (58.0% to 60.5%), and delayed patients wait about 46 minutes on average against 20 for those seen on time. Because delays do not track patient volume, the best opportunity lies in improving triage, staffing and patient flow rather than adding capacity alone. Reducing delays, fast-tracking primary-care cases and tracking satisfaction more completely should be the priorities for the coming months.

Source: Hospital Emergency Room Dashboard workbook (Excel data model), 9,216 patient records, April 2023 to October 2024.